

Weibull Distribution

①

A random Variable 'x' is said to have a Weibull distribution with parameters 'a' and 'b' if its density function is given by.

$$f(x) = \frac{ab}{a^b} x^{b-1} e^{-ax^b} \quad 0 < x < \infty, a, b > 0$$

Question

Show that in Weibull distribution.

$$\mu'_r = \left(\frac{1}{a}\right)^{\frac{r}{b}} \left[1 + \frac{r}{b}\right]$$

use it to find mean and Variance.

Sol

By definition

$$\mu'_r = E(x^r) = \int_0^{\infty} x^r f(x) dx$$

$$= ab \int_0^{\infty} x^r \cdot \frac{1}{a^b} x^{b-1} e^{-ax^b} dx$$

$$= ab \int_0^{\infty} x^{r+b-1} e^{-ax^b} dx$$

$$\text{put } ax^b = y$$

$$x^b = \frac{y}{a}$$

$$x = \left(\frac{y}{a}\right)^{\frac{1}{b}}$$

$$dx = \frac{1}{b} \left(\frac{y}{a}\right)^{\frac{1}{b}-1} \cdot \frac{1}{a} dy$$

$$\mu'_r = ab \int_0^{\infty} \left[\left(\frac{y}{a}\right)^{\frac{1}{b}}\right]^{r+b-1} e^{-y} \frac{1}{ab} \left(\frac{y}{a}\right)^{\frac{1}{b}-1} dy$$

$$\mu'_1 = \int_0^{\infty} \left(\frac{y}{a}\right)^{\frac{1}{b}} \left(\frac{y}{a}\right)^{\frac{1}{b}-1} e^{-y} \left(\frac{y}{a}\right)^{\frac{1}{b}-1} dy \quad (2)$$

$$= \int_0^{\infty} \left(\frac{y}{a}\right)^{\frac{1}{b}} e^{-y} dy$$

$$= \left(\frac{1}{a}\right)^{\frac{1}{b}} \int_0^{\infty} y^{\frac{1}{b}} e^{-y} dy$$

$$= \left(\frac{1}{a}\right)^{\frac{1}{b}} \int_0^{\infty} y^{\left(\frac{1}{b}+1\right)-1} e^{-y} dy$$

$$\mu'_1 = \left(\frac{1}{a}\right)^{\frac{1}{b}} \left[\frac{y^{\frac{1}{b}+1}}{\frac{1}{b}+1}\right]_0^{\infty}$$

put $x=1, 2$

$$\mu'_1 = \left(\frac{1}{a}\right)^{\frac{1}{b}} \left[\frac{1^{\frac{1}{b}+1}}{\frac{1}{b}+1}\right]$$

$$\mu'_2 = \left(\frac{1}{a}\right)^{\frac{1}{b}} \left[\frac{2^{\frac{1}{b}+1}}{\frac{1}{b}+1}\right]$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$= \left(\frac{1}{a}\right)^{\frac{1}{b}} \left[\frac{2^{\frac{1}{b}+1}}{\frac{1}{b}+1} - \left(\frac{1}{a}\right)^{\frac{1}{b}} \left[\frac{1^{\frac{1}{b}+1}}{\frac{1}{b}+1}\right]^2\right]$$

$$= \left(\frac{1}{a}\right)^{\frac{2}{b}} \left[\frac{2^{\frac{1}{b}+1}}{\frac{1}{b}+1} - \left(\frac{1}{a}\right)^{\frac{1}{b}} \left(\frac{1^{\frac{1}{b}+1}}{\frac{1}{b}+1}\right)^2\right]$$

$$\mu_2 = \left(\frac{1}{a}\right)^{\frac{2}{b}} \left[\frac{2^{\frac{1}{b}+1}}{\frac{1}{b}+1} - \left(\frac{1}{a}\right)^{\frac{1}{b}} \left(\frac{1^{\frac{1}{b}+1}}{\frac{1}{b}+1}\right)^2\right]$$

11 Pareto Distribution

①

The probability density function of the Pareto distribution is.

$$f(x; k, \alpha) = \begin{cases} \frac{\alpha k^\alpha}{x^{\alpha+1}} & \alpha > 0, \frac{x \geq k > 0}{k \leq x < \infty} \\ 0 & \text{otherwise} \end{cases}$$

Question

Find the r th moment about origin (μ'_r) of the Pareto distribution. Use it to find the Mean and Variance.

Sol By definition.

$$\begin{aligned} \mu'_r = E(x^r) &= \int_K^\infty x^r f(x) dx \\ &= \int_K^\infty x^r \cdot \frac{\alpha k^\alpha}{x^{\alpha+1}} dx \\ &= \alpha k^\alpha \int_K^\infty x^r \cdot x^{-\alpha-1} dx \\ &= \alpha k^\alpha \int_K^\infty x^{r-\alpha-1} dx \\ &= \alpha k^\alpha \left[\frac{x^{r-\alpha}}{r-\alpha} \right]_K^\infty \\ &= \frac{\alpha k^\alpha}{r-\alpha} \left[(\infty)^{r-\alpha} - K^{r-\alpha} \right] \\ &= -\alpha k^\alpha \cdot K^r \cdot K^{-\alpha} / (r-\alpha) \end{aligned}$$

$$\begin{aligned} \frac{1}{(\alpha)^{\alpha-r}} \\ \frac{1}{\infty} &= 0 \end{aligned}$$

$$\mu'_r = \frac{\alpha k^r}{\alpha - r} \quad \alpha > r \quad r \geq 1$$

put $r=1, 2$

$$\mu'_1 = \frac{\alpha k}{\alpha - 1}$$

$$\mu'_2 = \frac{\alpha k^2}{\alpha - 2}$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$= \frac{\alpha k^2}{\alpha - 2} - \left(\frac{\alpha k}{\alpha - 1} \right)^2$$

$$= \alpha k^2 \left[\frac{1}{\alpha - 2} - \frac{\alpha}{(\alpha - 1)^2} \right]$$

$$= \alpha k^2 \left[\frac{(\alpha - 1)^2 - \alpha(\alpha - 2)}{(\alpha - 2)(\alpha - 1)^2} \right]$$

$$= \alpha k^2 \left[\frac{\cancel{\alpha^2} + 1 - 2\cancel{\alpha} - \cancel{\alpha^2} + 2\cancel{\alpha}}{(\alpha - 2)(\alpha - 1)^2} \right]$$

$$= \frac{\alpha k^2}{(\alpha - 2)(\alpha - 1)^2} \quad \alpha > 2$$